Labs

Questions

1. What is Maven, and what role does it play in Java project management?
2. How are dependencies defined in a Maven `pom.xml` file? Provide an example of adding a dependency to a project.
3. Explain the concepts of lifecycles, phases, and goals in the Maven build process.
4. What is a fat JAR, and why is it beneficial for production use?
5. Is it advisable to run a project using `mvn spring-boot:run` in a production environment? Why or why not?
6. How does Spring Boot acquire configuration settings? Describe its hierarchical approach.
7. What is Project Lombok, and how does it simplify Java development?
8. What is inversion of control (IoC), and how does Spring implement it?
9. What is a bean in Spring? What are the two most common ways to define beans? Explain the purpose of the `@Primary` and `@Qualifier` annotations.
10. What is the role of application properties in configuring a Spring Boot application and its profiles? What is the key advantage of using properties instead of hardcoded configurations? How do these advantages align with the 12-Factor App methodology?
11. How does a logging system work, and how can log levels be used effectively during both development and production?
12. How do Spring Boot Actuators help in managing a microservice in a production environment?
13. What are the key features of Spring Data JPA?